SRM INSTITUTE OF SCIENCE AND TECHNOLOGY
DEPARTMENT OF MATHEMATICS
21MAB204T - PROBABILITY AND QUEUEING THEORY
Academic Year: 2025 - 2026 (Even Semester)

UNIT - IV : TUTORIAL SHEET – 11

S. No.	Problems
1.	A supermarket has two girls running up sales at the counters. If the service time for each customer is exponential with mean 4 minutes, and if people arrive in a Poisson fashion at the rate of 10 per hour. (i) What is the probability of having to wait for service? (ii) What is the expected percentage of idle time for each girl? (iii) If a customer has to wait, what is the expected length of waiting time?
2.	A petrol pump has 4 pumps. The service time follows an exponential distribution with mean of 6 minutes and cars arrive for service in a poisson process at the rate of 30 cars per hour. (i) What is the probability that an arrival will have to wait in the line. (ii) Find the average waiting time in the queue. (iii) Find the average time spent in the system. (iv) Find the average number of cars in the system.
3.	There are 3 typists in an office. Each typist can type an average of 6 letters per hour. If letters arrive for being typed at the rate of 15 letters per hour. (i) What fraction of time all the typist will be busy? (ii) What is the average number of letters waiting to be typed? (iii) What is the average time a letter has to spend waiting and for being typed?
4.	A petroleum company is considering expansion of its one unloading facility at its refinery. Due to random variations in weather, loading delays and other factors, ships arriving at the refinery to unload crude oil arrive at a rate of 5 ship per week. The service rate is 10 ships per week. Assume arrivals follow a Poisson process and the service rate is 10 ships per week. Assume arrivals follow a Poisson process and the service time exponential. (i) Find the average time a ship must wait before beginning to deliver its cargo to the refinery. (ii) If a second berth is rented, what will be the average number of ships waiting before being unloaded? (iii) What would be the average time a ship would wait before being unloaded with two berths? (iv) What is the average number of idle berths at any specified time?
5.	A library want to improve the service facilities in terms of the waiting time of its, borrowers. The library has two counters at present and borrowers arrive according to a Poisson distribution with arrival rate 1 at every 6 minutes and service time follows exponential distribution with a mean of 10 minutes. The library has relaxed its membership rules and a substantial increase in the number of borrrowers is expected. Find the number of additional counters to be provided if the arrival rate is expected to be twice the present value and the average waiting time of the borrower must be limited to half the present value.
6.	A tax consulting firm has three counters in its office to receive the people who have problems concerning their income and sales taxes. On the average 48 persons arrive in an 8 hour day. Each tax advisor spends 15 minutes on an average on an arrival. If the arrivals are Poisson distributed and service times are according to exponential, find the following: (i) Average number of customers in the system. (ii) Average number of customer waiting to be served. (iii) Average time a customer spend in the system. (iv) Average time a customer spend in the queue. (v) The number of hours each week a tax advisor spends performing his job.